

CHEM 3610 (Rowley) Midterm Exam 4 Equations and Constants

Gas Constant Values			
8.314	$\frac{J}{mole\ K}$	0.08314	$\frac{L\ bar}{mole\ K}$
0.08206	$\frac{L\ atm}{mole\ K}$	8.314	$\frac{m^3\ Pa}{mole\ K}$
Boltzmann Constant Values			
1.381×10^{-23}	$\frac{J}{K}$	0.6950	$\frac{cm^{-1}}{K}$

Conversions		
1 L atm	=	101.325 J
1 atm	=	1.01325 bar
1 atm	=	760 torr
1 atm	=	101,325 Pa

$$v = \frac{d[A]}{\nu_A dt} = \frac{1}{V} \frac{d\xi}{dt}$$

$$\ln\left(\frac{v_2}{v_1}\right) = m \ln\left(\frac{[A]_2}{[A]_1}\right)$$

$$\chi = \chi_0 e^{-t/\tau}$$

$$\tau = \frac{1}{k_r + k'_r}$$

$$k = A e^{-\frac{E_a}{RT}}$$

$$\ln\left(\frac{k_2}{k_1}\right) = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$$v_{\text{Lind.-Hinsh.}} = \frac{k_a k_b [A]^2}{k_b + k'_a [A]}$$

$$\tau_0 = \frac{1}{k_F + k_{IC} + k_{ISC}}$$

$$\phi_{F,0} = \frac{k_F}{k_F + k_{IC} + k_{ISC}} = k_F \tau$$

$$\frac{\phi_0}{\phi} = 1 + \tau_0 k_Q [Q]$$

$$\eta_T = 1 - \frac{\phi_F}{\phi_{F,0}} = \frac{R_0^6}{R_0^6 + R^6}$$

$$\frac{1}{v} = \frac{1}{v_{max}} + \left(\frac{K_M}{v_{max}}\right) \frac{1}{[S]_0}$$

$$v = P \sigma v_{rel} N_A^2 e^{-\frac{E_a}{RT}} [A][B]$$

$$k_d = \frac{8RT}{3\eta}$$

$$k \propto e^{\Delta S^\ddagger/R} e^{\Delta H^\ddagger/RT}$$

Periodic Table of the Elements

1 H Hydrogen 1.008	2 He Helium 4.003																	18 VIII A 8A
3 Li Lithium 6.941	4 Be Beryllium 9.012																	17 VII A 7A
11 Na Sodium 22.990	12 Mg Magnesium 24.305																	16 VI A 6A
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.631	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798	15 V A 5A
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.906	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.414	49 In Indium 114.818	50 Sn Tin 118.711	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.904	54 Xe Xenon 131.294	14 IV A 4A
55 Cs Cesium 132.905	56 Ba Barium 137.328	57 La Lanthanum 138.905	72 Hf Hafnium 178.49	73 Ta Tantalum 180.948	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.085	79 Au Gold 196.967	80 Hg Mercury 200.592	81 Tl Thallium 204.383	82 Pb Lead 207.2	83 Bi Bismuth 208.980	84 Po Polonium [208.982]	85 At Astatine 209.987	86 Rn Radon 222.018	13 III A 3A
87 Fr Francium 223.020	88 Ra Radium 226.025	89 Ac Actinium 227.028	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [278]	110 Ds Darmstadtium [281]	111 Rg Roentgenium [280]	112 Cn Copernicium [285]	113 Nh Nihonium [286]	114 Fl Flerovium [289]	115 Mc Moscovium [289]	116 Lv Livermorium [293]	117 Ts Tennessine [294]	118 Og Oganesson [294]	12 IIB 2B
																		11 IB 1B
																		10 VIII 8
																		9 VII B 7B
																		8 VI B 6B
																		7 V B 5B
																		6 IV B 4B
																		5 III B 3B
																		4 II B 2B
																		3 I B 1A
																		2 II A 2A
																		1 I A 1A

Lanthanide Series

Actinide Series